

AGUTS

Ahmedabad Gandhinagar Unified Transit System

Minimal Set of Functional Dependencies

Minimal Set of Functional Dependencies (Corrected v2)

★-marked rows reflect corrections identified by comparing against the reference schema.

Two relations (INTERCHANGE, SMART_CARD) now have two candidate keys each, noted explicitly.

Removed: INTERCHANGE_STOP junction table (replaced by 1:1 FK on INTERCHANGE.stop_id).

Summary Table

#	Relation	Candidate Key(s)	Minimal FD Set (condensed)
1	CITY	{city_id}	city_id → city_name, city_state
2	TRANSIT_OPERATOR	{operator_id}	operator_id → operator_name, service_type, city_id
3	METRO_LINE	{line_id}	line_id → line_name, line_code, operator_id
4	FARE_ZONE	{zone_id}	zone_id → zone_name, operator_id ★ operator_id added
5	DEPOT	{depot_id}	depot_id → depot_name, address, operator_id, city_id ★ city_id added
6	VEHICLE	{vehicle_id}	vehicle_id → registration_no, capacity, manufacture_year, manufacturer_name, manufacturer_info, status, operator_id, depot_id
7	BUS	{vehicle_id}	vehicle_id → fuel_type, bus_type
8	METRO_TRAIN	{vehicle_id}	vehicle_id → rake_number, num_coaches ★ train_number→rake_number
9	MAINTENANCE	{maintenance_id}	maintenance_id → vehicle_id, start_date, end_date, maintenance_type, description ★ maintenance_type added
10	STOP	{stop_id}	stop_id → stop_name, stop_type, latitude, longitude, city_id ★ zone_id removed
11	INTERCHANGE	{interchange_id}, {stop_id}	interchange_id → stop_id, interchange_type, description stop_id → interchange_id, interchange_type, description ★ 2 candidate keys (1:1 with STOP)

12	ROUTE	{route_id}	route_id → route_number, route_name, distance_km, operator_id, line_id ★ operator_id + line_id added
13	ROUTE_STOP	{route_id, stop_id, sequence_no}	{route_id, stop_id, sequence_no} → zone_id ★ PK extended; stop_id now PK+FK; zone_id added
14	STAFF	{staff_id}	staff_id → first_name, middle_name, last_name, role, contact, date_of_joining, operator_id, depot_id ★ name split; certificate removed
15	STAFF_CERTIFICATION	{staff_id, certification}	{staff_id, certification} → issued_date, expiry_date ★ NEW (MV attribute)
16	DRIVER	{staff_id}	staff_id → license_no
17	CONDUCTOR	{staff_id}	staff_id → badge_no
18	MOTORMAN	{staff_id}	staff_id → metro_license_no
19	TRIP	{trip_id}	trip_id → trip_date, trip_status, route_id, vehicle_id, staff_id ★ etd/eta/delay_minutes removed (derived)
20	TRIP_STOP	{trip_id, sequence_no}	{trip_id, sequence_no} → stop_id, sched_arrival, actual_arrival ★ stop_order→sequence_no
21	PASSENGER	{passenger_id}	passenger_id → first_name, middle_name, last_name, email, date_of_birth, concession_type, city_id ★ name split; phone removed
22	PASSENGER_PHONE	{passenger_id, phone}	{passenger_id, phone} → phone_type ★ NEW (MV attribute)
23	SMART_CARD	{card_id}, {passenger_id}	card_id → passenger_id, balance, issue_date, expiry_date, card_status passenger_id → card_id (2nd candidate key via UNIQUE)
24	PASS	{pass_id}	pass_id → pass_type, valid_from, valid_to, amount_paid, passenger_id, operator_id ★ fare_id→operator_id; amount_paid added
25	FARE	{fare_id}	fare_id → fare_amount, concession_type, operator_id, from_zone_id, to_zone_id ★ operator_id added; amount→fare_amount
26	TICKET_TRANSACTION	{txn_id}	txn_id → ticket_type, amount, txn_datetime, payment_mode, passenger_id, trip_id, card_id
27	COMPLAINT	{complaint_id}	complaint_id → category, description, status, filed_date, resolved_date, passenger_id, trip_id

Expanded Minimal FDs — Single-Attribute RHS Form

R1. CITY

Candidate Key(s): {city_id}

```
city_id → city_name  
city_id → city_state
```

R2. TRANSIT_OPERATOR

Candidate Key(s): {operator_id}

```
operator_id → operator_name  
operator_id → service_type  
operator_id → city_id
```

R3. METRO_LINE

Candidate Key(s): {line_id}

```
line_id → line_name  
line_id → line_code  
line_id → operator_id
```

R4. FARE_ZONE ★

Candidate Key(s): {zone_id}

```
zone_id → zone_name  
zone_id → operator_id
```

R5. DEPOT ★

Candidate Key(s): {depot_id}

```
depot_id → depot_name  
depot_id → address  
depot_id → operator_id  
depot_id → city_id
```

R6. VEHICLE

Candidate Key(s): {vehicle_id}

```
vehicle_id → registration_no  
vehicle_id → capacity  
vehicle_id → manufacture_year  
vehicle_id → manufacturer_name  
vehicle_id → manufacturer_info
```

```
vehicle_id → status  
vehicle_id → operator_id  
vehicle_id → depot_id
```

R7. BUS

Candidate Key(s): {vehicle_id}

```
vehicle_id → fuel_type  
vehicle_id → bus_type
```

R8. METRO_TRAIN ★

Candidate Key(s): {vehicle_id}

```
vehicle_id → rake_number  
vehicle_id → num_coaches
```

R9. MAINTENANCE ★

Candidate Key(s): {maintenance_id}

```
maintenance_id → vehicle_id  
maintenance_id → start_date  
maintenance_id → end_date  
maintenance_id → maintenance_type  
maintenance_id → description
```

R10. STOP ★

Candidate Key(s): {stop_id}

```
stop_id → stop_name  
stop_id → stop_type  
stop_id → latitude  
stop_id → longitude  
stop_id → city_id
```

R11. INTERCHANGE ★

Candidate Key(s): {interchange_id}, {stop_id}

```
interchange_id → stop_id  
interchange_id → interchange_type  
interchange_id → description  
stop_id → interchange_id (2nd candidate key)  
stop_id → interchange_type  
stop_id → description
```

R12. ROUTE ★

Candidate Key(s): {route_id}

```
route_id → route_number  
route_id → route_name  
route_id → distance_km  
route_id → operator_id  
route_id → line_id
```

R13. ROUTE_STOP ★

Candidate Key(s): {route_id, stop_id, sequence_no}

```
{route_id, stop_id, sequence_no} → zone_id
```

R14. STAFF ★

Candidate Key(s): {staff_id}

```
staff_id → first_name  
staff_id → middle_name  
staff_id → last_name  
staff_id → role  
staff_id → contact  
staff_id → date_of_joining  
staff_id → operator_id  
staff_id → depot_id
```

R15. STAFF_CERTIFICATION ★

Candidate Key(s): {staff_id, certification}

```
{staff_id, certification} → issued_date  
{staff_id, certification} → expiry_date
```

R16. DRIVER

Candidate Key(s): {staff_id}

```
staff_id → license_no
```

R17. CONDUCTOR

Candidate Key(s): {staff_id}

```
staff_id → badge_no
```

R18. MOTORMAN

Candidate Key(s): {staff_id}

```
staff_id → metro_license_no
```

R19. TRIP ★

Candidate Key(s): {trip_id}

```
trip_id → trip_date  
trip_id → trip_status  
trip_id → route_id  
trip_id → vehicle_id  
trip_id → staff_id
```

R20. TRIP_STOP ★

Candidate Key(s): {trip_id, sequence_no}

```
{trip_id, sequence_no} → stop_id  
{trip_id, sequence_no} → sched_arrival  
{trip_id, sequence_no} → actual_arrival
```

R21. PASSENGER ★

Candidate Key(s): {passenger_id}

```
passenger_id → first_name  
passenger_id → middle_name  
passenger_id → last_name  
passenger_id → email  
passenger_id → date_of_birth  
passenger_id → concession_type  
passenger_id → city_id
```

R22. PASSENGER_PHONE ★

Candidate Key(s): {passenger_id, phone}

```
{passenger_id, phone} → phone_type
```

R23. SMART_CARD

Candidate Key(s): {card_id}, {passenger_id}

```
card_id → passenger_id  
card_id → balance  
card_id → issue_date  
card_id → expiry_date  
card_id → card_status  
passenger_id → card_id (2nd CK)
```

R24. PASS ★

Candidate Key(s): {pass_id}

```
pass_id → pass_type  
pass_id → valid_from  
pass_id → valid_to  
pass_id → amount_paid  
pass_id → passenger_id  
pass_id → operator_id
```

R25. FARE ★

Candidate Key(s): {fare_id}

```
fare_id → fare_amount  
fare_id → concession_type  
fare_id → operator_id  
fare_id → from_zone_id  
fare_id → to_zone_id
```

R26. TICKET_TRANSACTION

Candidate Key(s): {txn_id}

```
txn_id → ticket_type  
txn_id → amount  
txn_id → txn_datetime  
txn_id → payment_mode  
txn_id → passenger_id  
txn_id → trip_id  
txn_id → card_id
```

R27. COMPLAINT

Candidate Key(s): {complaint_id}

```
complaint_id → category  
complaint_id → description  
complaint_id → status  
complaint_id → filed_date  
complaint_id → resolved_date  
complaint_id → passenger_id  
complaint_id → trip_id
```